SOLUTION KEY
DERYA
ALTINTAN

T.A.U. Engineering

	DIFFERENTIAL EQUATIONS Final		
Course : *Differential Equations*	Surname :	Department:	
Year : *2025-2026*	Name :		
Semester : *Fall*	Number :		
Lecturer : *Derya Altıntan*	Signature :		
Date : *9 January 2026*			
Time : *10:00*	5 QUESTION 100 POINTS		
Duration : *120 minutes*			

1	2	3	4	5	6	7	TOTAL	

1.(20 pts)Obtain the general solution of the following nonhomogeneous linear differential equation by using the method of undetermined coefficients

$$y'' - y' - 2y = e^{2x}.$$

y_h $r^2 - r - 2 = 0$ $(r-2)(r+1) = 0$ $r = 2$ $r = -1$

$$y_h = c_1 e^{2x} + c_2 e^{-x}$$

y_p $y_p = Axe^{2x}$

$$y_p' = Ae^{2x} + 2Axe^{2x}$$

$$y_p'' = 2Ae^{2x} + 2A(e^{2x} + 2xe^{2x})$$

$$= 4Ae^{2x} + 4Axe^{2x}$$

$$4Ae^{2x} + 4Axe^{2x} - (Ae^{2x} + 2Axe^{2x}) - 2Axe^{2x} = e^{2x}$$

$$3Ae^{2x} = e^{2x} \qquad A = \frac{1}{3} \qquad y_p = \frac{1}{3}xe^{2x}$$

$$y(x) = c_1 e^{2x} + c_2 e^{-x} + \frac{1}{3}xe^{2x}$$

2.(15+5 pts) a) Solve the following problems using the Laplace transform:

$$y'' + 2y' + y = \delta(t-1) + u_3(t), \qquad y(0) = 0, \quad y'(0) = 0.$$

Not: $\delta(t-a)$ Dirac delta, $u_a(t)$ Heaviside.

$$\mathcal{L}\{y''\} + 2\mathcal{L}\{y'\} + \mathcal{L}\{y\} = \mathcal{L}\{\delta(t-1)\} + \mathcal{L}\{u_3(t)\} \qquad e^{-s}, \quad e^{-3s}/s$$

$$s^2 Y(s) - s\,y(0) - y'(0) + 2sY(s) - 2y(0) + Y(s) = e^{-s} + \frac{e^{-3s}}{s}$$

$$(s^2+2s+1)\,Y(s) = e^{-s} + \frac{e^{-3s}}{s} \qquad Y(s) = \frac{e^{-s}}{(s+1)^2} + \frac{e^{-3s}}{s(s+1)^2}$$

$$\frac{1}{s(s+1)^2} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{(s+1)^2} = \frac{A(s+1)^2 + B(s+1)s + Cs}{(s+1)^2 s} = \frac{(A+B)s^2 + (2A+B+C)s + A}{(s+1)^2 s}$$

$A=1, B=-1, C=-1$

$$Y(s) = \frac{e^{-s}}{(s+1)^2} + e^{-3s}\left(\frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2}\right)$$

$$\mathcal{L}^{-1}\left\{\frac{e^{-s}}{(s+1)^2}\right\} = u_1(t)\cdot f(t-1) \quad \left[\mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2}\right\} = e^{-t}\cdot t\right] \qquad e^{-(t-1)}(t-1)$$

$$\mathcal{L}^{-1}\left\{\frac{e^{-3s}}{(s+1)^2}\right\} = u_3(t)\,f(t-3) = u_3(t)\,e^{-(t-3)}(t-3)$$

$$\mathcal{L}^{-1}\left\{\frac{e^{-3s}}{s}\right\} = u_3(t)\,f(t-3) \quad \left[\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1\right] \qquad 1$$

$$\mathcal{L}^{-1}\left\{\frac{e^{-3s}}{s+1}\right\} = u_3(t)\,e^{-(t-3)} \quad \left[\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t}\right]$$

b) $y(x) = x^3 + \int_0^x \sin(x-t)y(t)\,dt$

FOR STUDENTS USING CONVOLUTION

$$\mathcal{L}\{y(x)\} = \mathcal{L}\{x^3\} + \mathcal{L}\left\{\int_0^x \sin(x-t)\cdot y(t)\,dt\right\}$$

$$= \mathcal{L}\{x^3\} + \underbrace{\mathcal{L}\{\sin x * y(x)\}}_{\mathcal{L}\{\sin x\}\,\mathcal{L}\{y(x)\}}$$

$$Y(s) = \underbrace{\frac{e^{-s}}{(s+1)^2}}_{u_1(t)\cdot f(t-1)} + \underbrace{e^{-3s}\frac{1}{s\cdot(s+1)^2}}_{F(s)}$$

$$F(s) = \frac{1}{s}\cdot\frac{1}{(s+1)^2} = \{1\} * \{t\cdot e^{-t}\}$$

$$\mathcal{L}\{1\}\quad \mathcal{L}\{t\cdot e^{-t}\}$$

$$1 * t\cdot e^{-t} = \int_0^t 1\cdot(t-\tau)\,e^{-(t-\tau)}\,d\tau$$

$$= \int_0^t \tau\cdot e^{-\tau}\,d\tau$$

$$= 1 - (t+1)\cdot e^{-t} = f(t)$$

$$\mathcal{L}^{-1}\{e^{-3s}F(s)\} = u_3(t)\cdot f(t-3)$$

$$= u_3(t)\left[1 - (t-2)e^{-(t-3)}\right]$$

$$\mathcal{L}\{y\} = \frac{\frac{3!}{s^4}}{\frac{s^2}{s^2+1}} = \frac{3!}{s^4}\cdot\frac{s^2+1}{s^2} = \frac{3!}{s^4} + \frac{3!}{s^6}\cdot\frac{5!}{5!}$$

$$= \frac{3!}{s^4} + \frac{5!}{s^6}\cdot\frac{1}{20}$$

$$\mathcal{L}\{y\} = \frac{3!}{s^4} + \frac{5!}{s^6}\cdot\frac{1}{20}$$

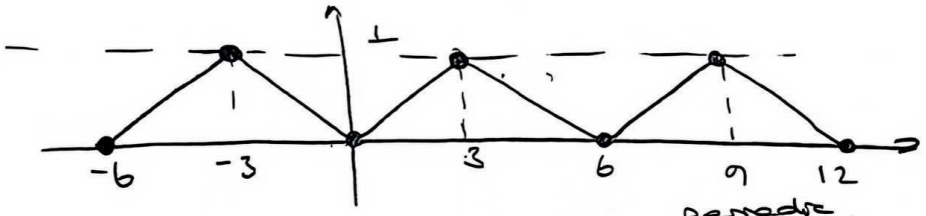
$$y(x) = x^3 + \frac{x^5}{20}$$

3.(20 pts) Let

$$f(t) = \frac{t}{3}, \qquad 0 \le t \le 3.$$

a) Consider the even periodic extension of f to $[-3,3]$ with period 6. (i) Find the Fourier series representation of the even periodic extension. (ii) Sketch the graph of the even periodic extension. (iii) Find $S_f(-1)$.

$$\tilde{f}(t) = \begin{cases} f(-t), & -3 < t < 0 \\ f(t), & 0 \le t \le 3. \end{cases} \Rightarrow \tilde{f} = \begin{cases} -t/3 & -3 < t < 0 \\ t/3 & 0 \le t \le 3. \end{cases}$$



Periodic extension. $T = 6$

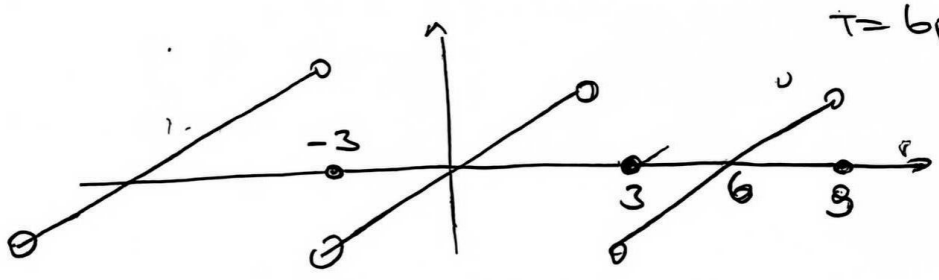
$S_f(-1) = 1/3.$

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b) Consider the odd periodic extension of f to $[-3,3]$ with period 6. (i) Find the Fourier series representation of the odd periodic extension. (ii) Sketch the graph of the odd periodic extension. (iii) Find $S_f(-1)$.

$$\tilde{f}(t) = \begin{cases} -f(-t) & -3 < t < 0 \\ 0 & t = 0 \; t = 3 \\ f(t) & 0 < t < 3 \end{cases} = \begin{cases} t/3 & -3 < t < 0 \\ 0 & t = 0, t = 3 \\ t/3 & 0 < t < 3 \end{cases}$$

$T = 6$ p. extension



$S_f(-1) = -1/3.$

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8. (20 pts) Find the solution to the heat flow problem

$$\frac{\partial u}{\partial t} = 5\frac{\partial^2 u}{\partial x^2}, \qquad 0 < x < \pi;\ t > 0,$$

with homogeneous Dirichlet boundary conditions

$$u(0,t) = u(\pi,t) = 0, \qquad t > 0,$$

and initial condition

$$u(x,0) = \sin(3x) - \sin(5x)$$

$u_{xx} = X''$ $\quad u_t = X\dot{T}$ $\quad X\dot{T} = 5X''T$ $\quad \frac{X''}{X} = \frac{1}{5}\frac{\dot{T}}{T} = -\lambda$

$X'' + \lambda X = 0$ $\quad \dot{T} + 5\lambda T = 0$

$u(0,t) = 0 \Rightarrow X(0)T(t) = 0 \quad X(\pi)T(t) = 0$

① $\lambda = 0$ $\quad r^2 = 0$ $\quad r_{1,2} = 0$ $\quad X(0) = 0$ $\quad X(\pi) = 0$

$X(x) = c_1 + c_2 x$ $\quad X(x) = 0$ trivial

$X(0) = c_1 = 0$

$X(\pi) = c_2 \cdot \pi = 0 \quad c_2 = 0$

F.S.S. $= \{1, x\}$.

② $\lambda < 0$ $\quad \lambda = -\mu^2$ $\quad X'' - \mu^2 X = 0$ $\quad$ F.S.S. $= \{e^{\mu x}, e^{-\mu x}\}$

$X(x) = c_1 e^{\mu x} + c_2 e^{-\mu x}$ $\quad X(0) = c_1 + c_2 = 0 \quad \Rightarrow c_1 = -c_2$

$X(\pi) = c_1 e^{\mu\pi} + c_2 e^{-\mu\pi} = 0 \Rightarrow c_1 e^{\mu\pi} - \frac{c_1}{e^{\mu\pi}} = 0 \quad c_1 = 0$

$X(x) = 0$ trivial sol.

③ $\lambda > 0$ $\quad \lambda = \mu^2$ $\quad X'' + \mu^2 X = 0$ $\quad$ F.S.S. $= \{\cos\mu x, \sin\mu x\}$

$X(x) = c_1 \cos\mu x + c_2 \sin\mu x$

$X(0) = c_1 = 0$

$X(\pi) = -c_2 \sin\mu x = 0. \quad c_2 \neq 0 \quad \mu = n$

$\lambda = n^2$

$X_n = \sin(nx)$

T_n $\quad \dot{T} + 5\lambda T = 0 \quad \frac{\dot{T}}{T} = -5\lambda \quad \ln T(t) = -5\lambda t \cdot \quad u_n(x,t) = X(x)T_n(t)$

$u_n(x,t) = e^{-5n^2 t} \cdot \sin(nx)$

$$u(x,0) = \sum_{n=1}^{\infty} c_n u_n(x,0) = \sum_{n=1}^{\infty} c_n \sin(nx) e^{-5n^2 t}$$

$u(x,t) = c_1 \sin x + c_2 \sin(2x) + c_3 \sin(3x) + \cdots = \sin(3x) - \sin 5x$

$c_n = \begin{cases} 1 & n = 3 \\ -1 & n = 5 \\ 0 & n \neq 3, 5 \end{cases}$

$u(x,t) = \sin(3x)e^{-45t} - \sin(5x)e^{-125t}$

5.(20 pts) Find a power series solution about $x = 0$ for a general solution to the differential equation

$$y'' - xy' + 4y = 0$$

$$y = \sum_{n=0}^{\infty} a_n x^n \qquad y' = \sum_{n=1}^{\infty} a_n n x^{n-1} \qquad y'' = \sum_{n=2}^{\infty} a_n (n-1) n x^{n-2}$$

$$\sum_{n=2}^{\infty} a_n (n-1) n x^{n-2} - \sum_{n=1}^{\infty} a_n \cdot x^n + \sum_{n=0}^{\infty} 4 \cdot a_n x^n = 0$$

$$\sum_{n=0}^{\infty} a_{n+2}(n+2)(n+1) x^n$$

$$a_2 + 4a_0 + \sum_{n=1}^{\infty} a_{n+2}(n+1)(n+2) x^n - \sum_{n=1}^{\infty} a_n n \cdot x^n + \sum_{n=1}^{\infty} 4 a_n x^n = 0$$

$$2a_2 + 4a_0 + \sum_{n=1}^{\infty} [a_{n+2}(n+1)(n+2) - a_n n + 4a_n] x^n = 0$$

$$2a_2 + 4a_0 = 0 \Rightarrow a_2 = -2a_0$$

$$a_{n+2} = \frac{n-4}{(n+1)(n+2)} a_n$$

$$a_4 = \frac{1}{3} a_0$$

$$a_6 = 0$$

$$a_8 = \frac{2}{8 \cdot 7} \cdot a_6 = 0$$

$$\Big\} \; a_{2n} = 0$$

$$a_{2n+1} = \frac{(2n-5)}{2n(2n+1)} a_{2n-1} = \frac{(2n-5)(2n-7)}{2n(2n+1)(2n-1)(2n-2)} a_{2n-3}$$

$$= \frac{(2n-5)(2n-7)(2n-9)}{2n(2n+1)(2n-1)(2n-2)(2n-3)(2n-4)} a_{2n-5} = \cdots = \frac{(2n-5)\cdots(-3)}{(2n+1)!} a_1$$

$$y(x) = a_0 \left[1 - 2x^2 + \frac{1}{3}x^4\right] + a_1 \left[x + \sum_{k=1}^{\infty} \frac{(2k-5)\cdots(-3)}{(2k+1)!}\right]$$

(3 c)

$$S_{\text{even}}(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi^2}\left[(-1)^n - 1\right]\cos\left(\frac{n\pi}{3}x\right)$$

(3 d)

$$S_{\text{odd}} = \sum_{n=1}^{\infty} \frac{2(1+(-1)^{n+1})}{n\pi}\sin\left(\frac{n\pi}{3}x\right)$$